F24-070-D-INSIGHTVISION

Project Team

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Chapter 1

Introduction

This proposal presents "Insight Vision," an AI-driven system created to streamline video review. Instead of manually watching hours of footage, the system identifies key events and summarizes pre-recorded videos. This makes post-event analysis faster and more efficient, saving time and effort in surveillance operations.

1.1 Problem Statement

The problem of inefficient manual review of pre-recorded CCTV footage affects security teams and organizations that rely on surveillance to ensure safety and monitor activities. The impact of the problem is time-consuming video analysis, missed critical events, and increased labor costs, which can lead to delayed responses to security threats. A successful solution would include automated event detection, reduced workload for security personnel, improved response times, and enhanced overall effectiveness of surveillance operations.

1.2 Scope

The scope of our project is to develop and design an AI system that automatically reviews pre-recorded video. Our system will detect motion, identify important events, label objects, and create brief summaries with timestamps. We will also use a large language model (LLM) so that the system can improve how well it recognizes actions and events.

The system will make video segments easy to navigate and provide a simple interface for quick analysis. It's mainly for indoor surveillance but can be adapted for other uses. The goal is to save time, cut down on manual work, and make surveillance more accurate.

1.3 Modules

1.3.1 Module 1: Event Detection and Analysis

This module focuses on detecting events, generating timestamps, recognizing actions, and tracking objects within the footage.

1. **Event-Based Video Chunking:**
    - This feature automatically divides long video footage into smaller, more manageable chunks.
    - It identifies any movement as an event, which is added to the environment for further analysis.
2. **Timestamp Generation for Key Events:**
    - This feature generates accurate timestamps for every detected action or movement.
    - Each significant event is marked with a timestamp for easy navigation during post-event review.
3. **Action Recognition and Labeling:**
    - Detects and labels movements in the footage based on object classification.
    - Differentiates between subjects like an intruder or a harmless entity such as a baby.
4. **Object Identification and Tracking:**
    - Tracks detected objects, such as people or animals, throughout the footage.
    - Records movements like walking, running, or minor motion and includes them in the summary.

1.3.2 Module 2: Summarization and Adaptive Reporting

This module handles the compilation of summarized videos, adapts to different environments for improved accuracy, and generates detailed event-based reports.

1. **Summarized Video Compilation:**
    - Compiles all detected events into a concise video summary.

- Allows users to view a condensed version of the footage, highlighting key moments and saving time.

2. **Event Identification and Highlighting:**

   - Identifies significant events within the footage.
   - Highlights these key moments to allow easier review and analysis.
   - "The literature review highlights that event detection is not merely about identifying actions but also involves understanding interactions with objects and other individuals." [1].

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

User Class	Description
Security personnel	Security personnel are the main users of the "Insight Vision" system. They monitor surveillance footage and respond to incidents. These trained professionals need a quick, efficient way to review footage. The system's video summary feature helps them focus on key events, saving time by skipping irrelevant content.
System Administrators	System administrators manage the "Insight Vision" system, overseeing user access, software updates, and performance. With technical expertise, they ensure seamless integration with existing surveillance technologies and adjust event detection accuracy as needed.
Testers	Individuals responsible for testing the application for bugs, usability issues, and ensuring the platform works as intended across different devices and environments.
End Users	End users, like managers or executives, occasionally review footage and need a simple interface to quickly access summarized videos or specific events. Their goal is to stay informed about relevant security incidents without needing extensive training.

Chapter 2

Project Requirements

2.1 Use Case 1: Upload Video for Event Detection

- **Scope**: Video Event Detection System
- **Level**: User goal
- **Primary Actor**: User (e.g., security team member)
- **Stakeholders and Interests**:
  - User: Wants to upload pre-recorded videos for event detection.
  - Security Team: Needs accurate and timely event detection.
  - System Administrator: Ensures smooth uploading and processing.
- **Preconditions**: User must be logged in, and the video file must be supported by the system.
- **Success Guarantee**: The video is uploaded successfully and prepared for analysis.

Main Success Scenario

1. User selects "Upload Video."
2. User browses for and selects the video file.
3. System verifies file format and size.
4. Video is uploaded and stored.
5. Analysis begins on the uploaded video.
6. The system confirms upload success.

Extensions

- Unsupported video format: User receives an error message.
- Upload failure due to network issues: User is prompted to retry.
- User cancels upload: System aborts the process.

Special Requirements

- Video upload must support formats like MP4, AVI, and MOV.
- Upload size limit is 1 GB.

Technology and Data Variations

- Web upload interfaces.

2.2 Use Case 2: Event-Based Video Chunking

- **Scope**: Video Processing Module
- **Level**: Subfunction
- **Primary Actor**: System (initiated by the user after video upload)
- **Stakeholders and Interests**:
  - User: Wants a more manageable, event-based summary of the video.
  - System Administrator: Ensures efficient processing of video chunks.
- **Preconditions**: Video must be uploaded and ready for processing.
- **Success Guarantee**: Video is automatically divided into event-based chunks.

Main Success Scenario

1. Video analysis begins automatically after upload.
2. The system identifies motion or significant events in the video.
3. The system creates individual video chunks based on detected events.
4. Chunks are saved for further analysis.

Extensions

- No detectable motion: System prompts user for manual review.
- If video is too large: System alerts the user about extended processing time.

Special Requirements

- System should chunk videos efficiently to minimize processing delays.

Technology and Data Variations

- Chunking algorithms for motion detection.

2.3 Use Case 3: Timestamp Generation for Key Events

- **Scope**: Video Processing Module
- **Level**: Subfunction
- **Primary Actor**: System
- **Stakeholders and Interests**:
  - User: Needs accurate timestamps for key events in video for easier review.
  - System Administrator: Ensures timestamp accuracy.
- **Preconditions**: Events must be detected in the uploaded video.
- **Success Guarantee**: Timestamps are generated for all detected events.

Main Success Scenario

1. System processes video chunks for events.
2. For each event, a timestamp is created.
3. Timestamps are saved along with video segments for user review.

Extensions

- Event duration too short: System adjusts timestamp to capture the full event.

Special Requirements

- Timestamps must be accurate to within 0.5 seconds of the detected event.

Technology and Data Variations

- System shows timestamp formats to users.

2.4 Use Case 4: Object Tracking

- **Scope**: Video Processing Module
- **Level**: Subfunction
- **Primary Actor**: System
- **Stakeholders and Interests**:
  - User: Needs to track movements of specific objects within the video.
  - System Administrator: Ensures efficient and accurate tracking.
- **Preconditions**: Video must be processed and objects identified.
- **Success Guarantee**: Objects are tracked throughout the video.

Main Success Scenario

1. System identifies objects in the video (e.g., person, vehicle).
2. System tracks each object's movements.
3. System records tracking data and associates it with video segments.

Extensions

- 2a. Multiple objects detected simultaneously: System tracks them individually.
- 2b. Object leaves the frame: System notifies user that object tracking ended.

Special Requirements

- The system must track objects with high accuracy.

Technology and Data Variations

- Object tracking algorithms vary depending on object type.

2.5 Use Case 5: Action Recognition and Labeling

- **Scope**: Video Processing Module
- **Level**: Subfunction
- **Primary Actor**: System
- **Stakeholders and Interests**:
  - User: Needs to understand what actions occurred in the video.
  - System Administrator: Ensures accuracy in action recognition.
- **Preconditions**: Video must be uploaded and processed.
- **Success Guarantee**: Actions are detected and labeled correctly.

Main Success Scenario

1. System analyzes video chunks for movements.
2. System detects specific actions (e.g., walking, running, sitting).
3. Each action is labeled in the video and associated with the relevant timestamps.

Extensions

- Unrecognizable actions: System flags for user review.
- Multiple actions simultaneously: System detects and labels each.

Special Requirements

- System must differentiate between distinct actions.

Technology and Data Variations

- Action recognition algorithms used.

2.6 Use Case 6: Generate Summarized Video

- **Scope**: Summarization and Reporting Module
- **Level**: User goal
- **Primary Actor**: User
- **Stakeholders and Interests**:
  - User: Wants a concise version of the original video highlighting key events.
  - System Administrator: Ensures proper video summarization.
- **Preconditions**: Video must be processed with key events and objects identified.
- **Success Guarantee**: Summarized video is generated and available for review.

Main Success Scenario

1. User selects the "Generate Summary" option.
2. System compiles key events and creates a summarized version.
3. System displays the summarized video for user review.

Extensions

- If summarization fails: User receives an error message and can retry.

Special Requirements

- The summarized video should be less that of the original length.

Technology and Data Variations

- Different summarization algorithms based on video length and complexity.

2.7 Event-Response Table for Backend Functionalities

This table outlines the various events in the system and how the backend processes respond to them.

Table 2.1: Event-Response Table for Backend Workflow

Event	System Response
User uploads video through the interface	The video is sent to the Video Processing Module for initial analysis.
Motion detected in the video	Motion Detection Module is triggered, identifying and marking areas of movement in the footage.
Object detected within the video	The system initiates Object Tracking to follow the movement of the identified object through the frames.
Recognized action (e.g., walking, running)	The Action Recognition module labels the identified action for use in the textual summary.
LLM generates textual summary for actions	The LLM produces a textual summary based on detected and recognized actions in the footage.
Footage chunked into smaller segments	Video Chunking Module breaks down long footage into manageable segments for better analysis.
LLM generates visual contextual captions for chunks	The LLM provides detailed captions explaining the content of each chunk based on detected events.
Processed results stored in the database	The system stores the summarized video and metadata (textual summary, object actions) in the database for future use.
User retrieves summarized video and report from the interface	The processed and summarized footage is made available to the user, along with its textual and visual summaries.

2.8 Functional Requirements

This section describes the functional requirements of the system expressed in natural language. The organization of functional requirements is by module and specific features within each module. These requirements detail the system's functionalities and expected behavior.

2.8.1 Module 1: Event Detection and Analysis

Following are the functional requirements for Module 1:

1. **FR1: Event Detection**

    - The system shall detect movements in the video footage and classify them as events.

- The system shall differentiate between significant and insignificant movements based on pre-defined parameters.[1].

2. **FR2: Timestamp Generation**

   - The system shall generate timestamps for each detected event.
   - Timestamps shall be accurate to within 1 second of the detected event.

3. **FR3: Action Recognition and Labeling**

   - The system shall recognize different actions based on object classification in the footage.
   - The system shall label identified actions (e.g., walking, running, sitting) appropriately.

4. **FR4: Object Tracking**

   - The system shall track moving objects within the video footage.
   - It shall maintain the identity of tracked objects throughout the footage.

2.8.2 Module 2: Summarization and Adaptive Reporting

Following are the functional requirements for Module 2:

1. **FR1: Summarized Video Compilation**

   - The system shall compile detected events into a summarized video.
   - The summary video shall condense footage based on key moments while maintaining the chronological order of events.

2. **FR3: Event Highlighting**

   - The system shall identify and highlight significant events during playback.
   - Highlighted events shall be visually distinguishable from other parts of the footage for easier review.

2.9 Non-Functional Requirements

This section specifies non-functional requirements for the system, focusing on aspects such as reliability, usability, performance, and security.

2.9.1 Reliability

Reliability refers to the consistency of the system in performing its intended functions without failure.

- REL-1: The system shall maintain a Mean Time Between Failures (MTBF) to ensure consistent performance.
- REL-2: In the event of a failure, the system shall provide error logs to assist in diagnosing the issue and facilitate recovery within 1 hour.

2.9.2 Usability

Usability is critical for ensuring that users can interact with the system effectively and efficiently.

- USE-1: Users shall be able to navigate the interface without prior training, with all functionalities accessible within 3 clicks.
- USE-2 The system shall provide contextual help and error messages to guide users in case of input errors.

2.9.3 Performance

Performance requirements specify the expected efficiency of the system during operations.

- PER-1: At least 95% of detected events shall be accurately identified within the first processing attempt.

2.9.4 Security

Security requirements are essential to protect user data and ensure system integrity.

- SEC-1: The system shall ensure that all video uploads are secured to prevent unauthorized access.
- SEC-2: User access to the system shall be restricted based on roles, ensuring that only authorized personnel can access sensitive data.
- SEC-3: The system shall log all access attempts and modifications to the database to provide a clear audit trail for security purposes.

2.10 Domain Model

The domain model for our project represents the key entities, attributes, and relationships involved in the system. The image below shows the visual representation of the domain model.

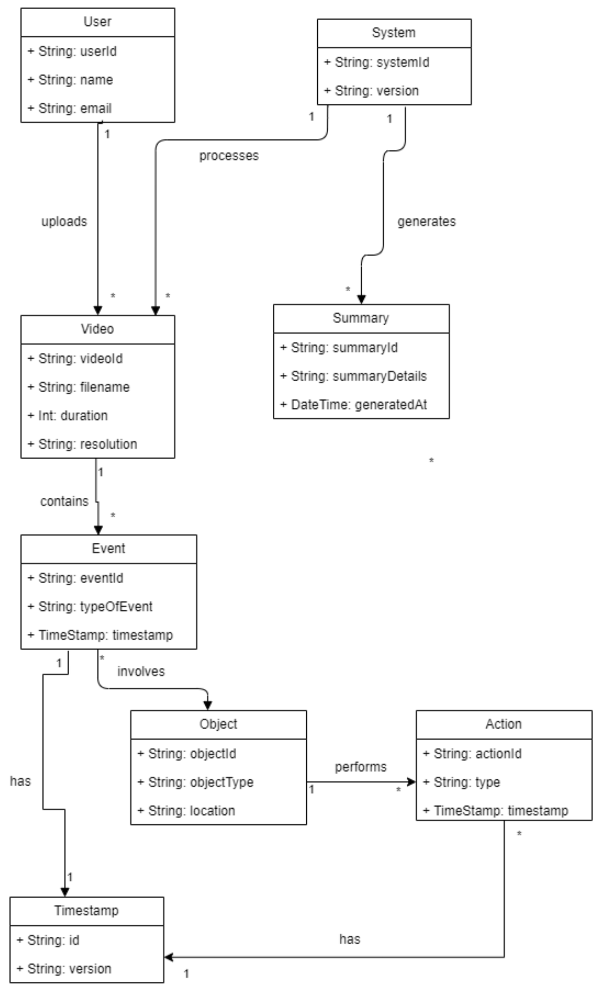

Figure 2.1: Domain Model Representation

Chapter 3

System Overview

3.1 Architectural Design

The architecture of the system is designed using a modular approach to achieve flexibility, scalability, and maintainability. The system consists of multiple components that work together to process video footage, detect events, and generate concise summaries.

The high-level system architecture is composed of the following key modules:

- **User Interface Module**: This module allows users to upload video footage and view the summarized results. It serves as the entry point for user interactions.

- **Video Processing Module**: Responsible for handling the uploaded videos, this module performs tasks such as video chunking, motion detection, and object tracking.

- **Event Detection Module**: This module identifies key events and actions in the video based on motion, object behavior, and other environmental cues.

- **Summarization Module**: This module compiles the detected events and actions into a concise video summary, highlighting the most relevant events for quick review.

- **LLM Integration Module**: The system incorporates a large language model (LLM) to generate text-based summaries and provide contextual captions based on the video content.

- **Database Module**: Stores the video data, events, and summaries for future reference and reporting purposes.

The system follows a layered architecture pattern, ensuring clear separation of concerns between the user interface, processing, and data storage. Each module interacts with other components through defined interfaces, ensuring modularity and ease of future extension.

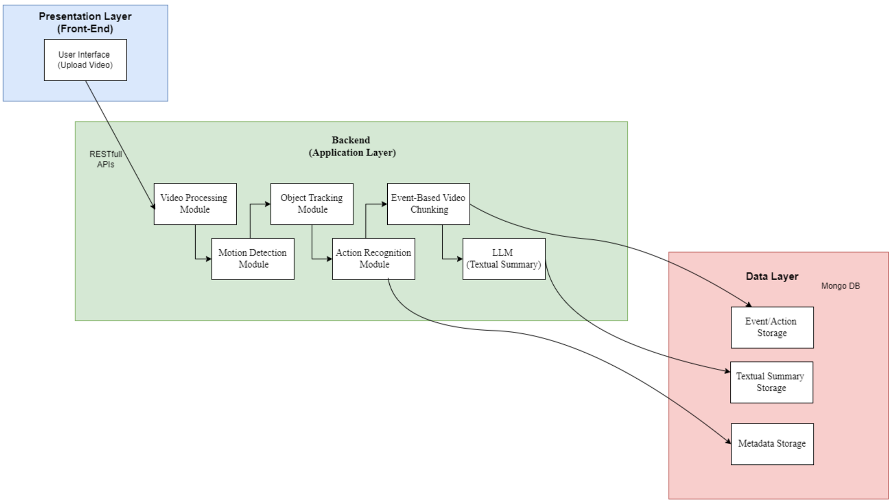

Figure 3.1: System Architecture Diagram

The architectural diagram (Figure 3.1) provides a visual representation of the system's key components and their relationships. It shows the interactions between the user interface, backend processing modules, and the database, emphasizing the flow of data and events through the system.

3.2 Design Models

This section presents the design models that provide an overview of the system structure and behavior using an Object-Oriented Development approach. Each diagram is accompanied by a brief description to explain its relevance and purpose in the system's design.

3.2.1 Class Diagram

The Class Diagram provides a static view of the system, illustrating the system's objects, attributes, methods, and the relationships between them. In the context of this project, the diagram includes classes such as 'User', 'Video', 'Event', 'Object', and 'System', showing how they interact.

Entities in the Class Diagram:

- **User**: Represents users interacting with the system, primarily uploading videos.
- **Video**: Stores the video data and its attributes such as duration and format.
- **Event**: Represents detected events in the video footage, linked with timestamps.
- **Object**: Represents the objects (e.g., people, animals) identified in the video.
- **Action**: Captures the actions (e.g., walking, running) performed by objects.
- **System**: Handles video processing, including event detection and summarization.
- **Summary**: Compiles the detected events and actions into a final summary.

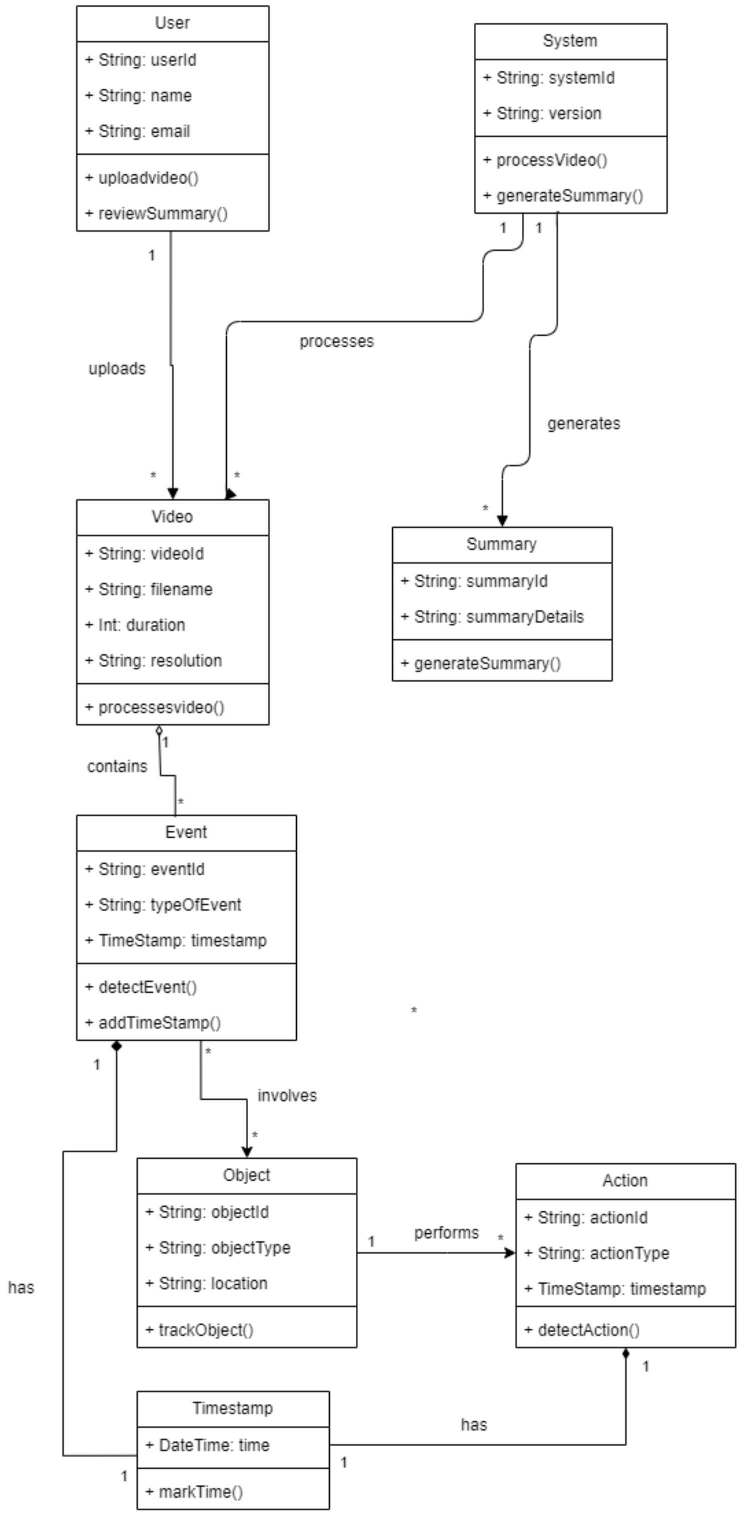

Figure 3.2: Class Diagram: Key Entities and Relationships in the System

3.2.2 Activity Diagrams

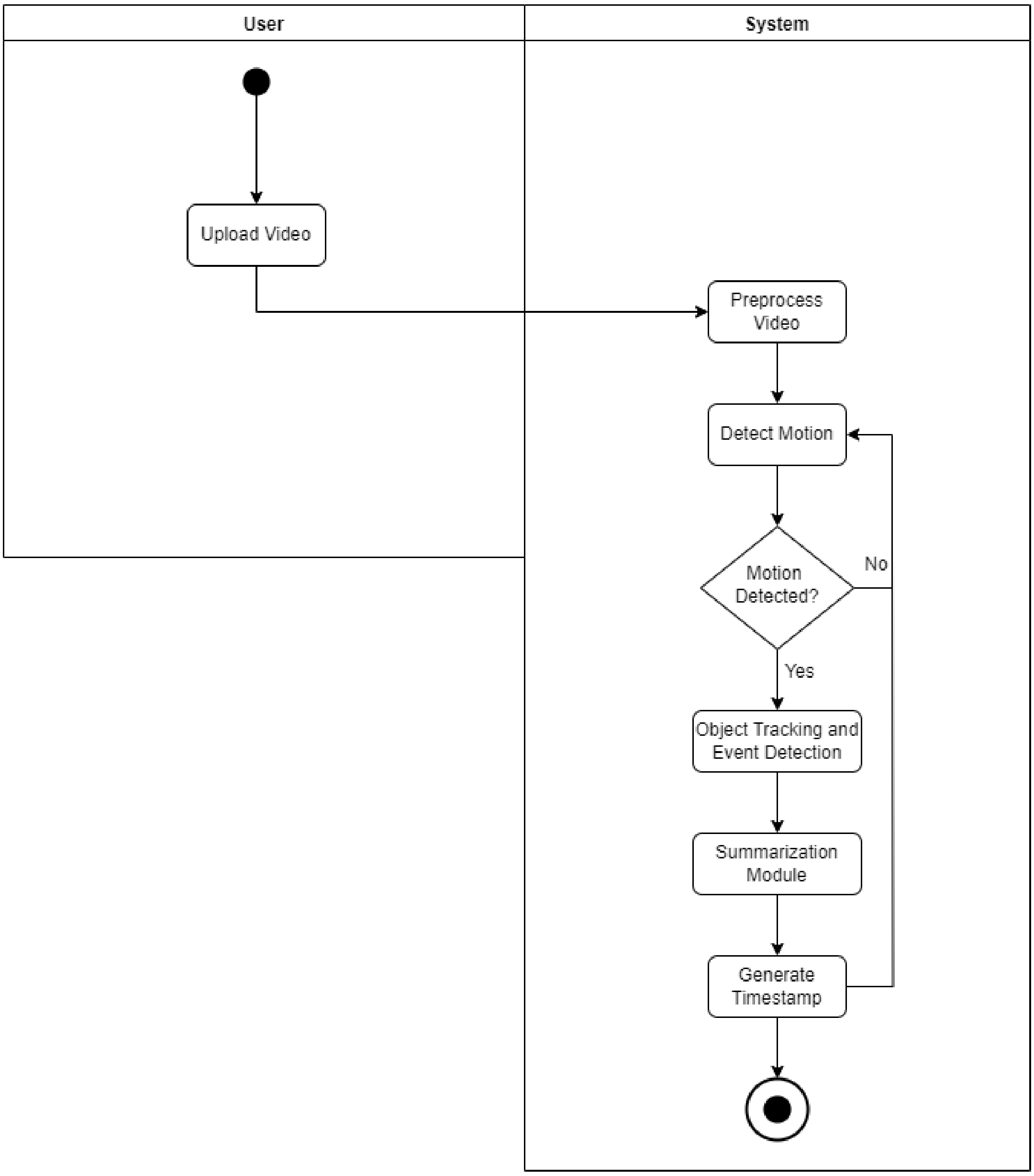

Figure 3.3: Activity Diagram 1: Video Upload and Processing Flow

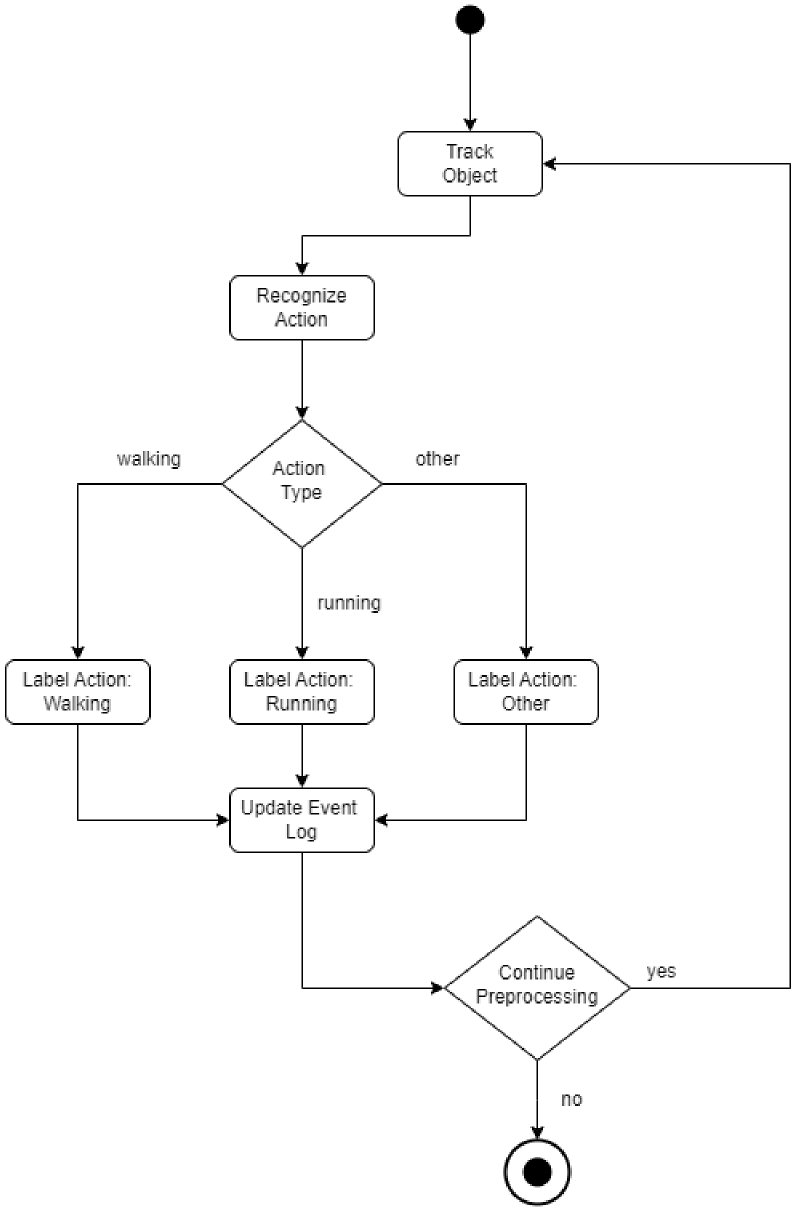

Figure 3.4: Activity Diagram 2: Object Tracking Process

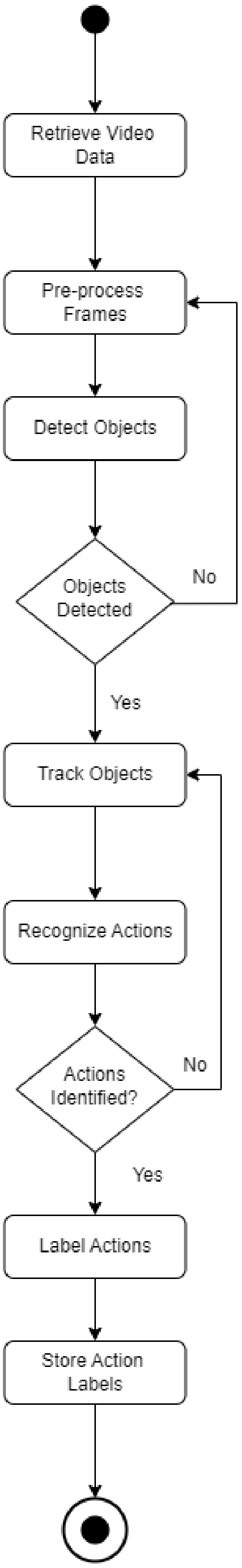

Figure 3.5: Activity Diagram 3: Action Recognition and Labeling

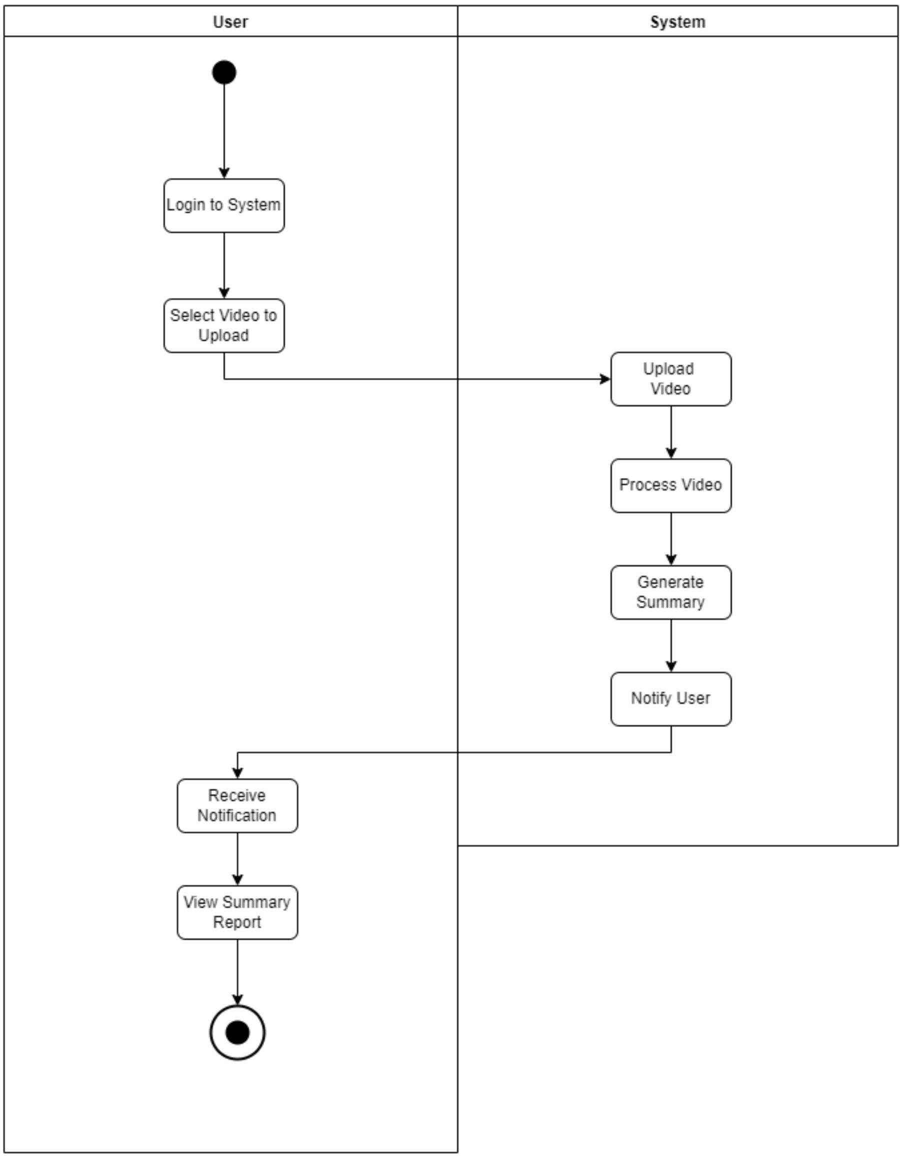

Figure 3.6: Activity Diagram 4: Video Upload & User Interaction

3.2.3 System Sequence Diagrams

3.2.3.1 Upload Video for Event Detection

The SSD for the "Upload Video for Event Detection" use case illustrates the interaction between the user and the system during the video upload process. The user initiates the upload, and the system processes the video for further analysis.

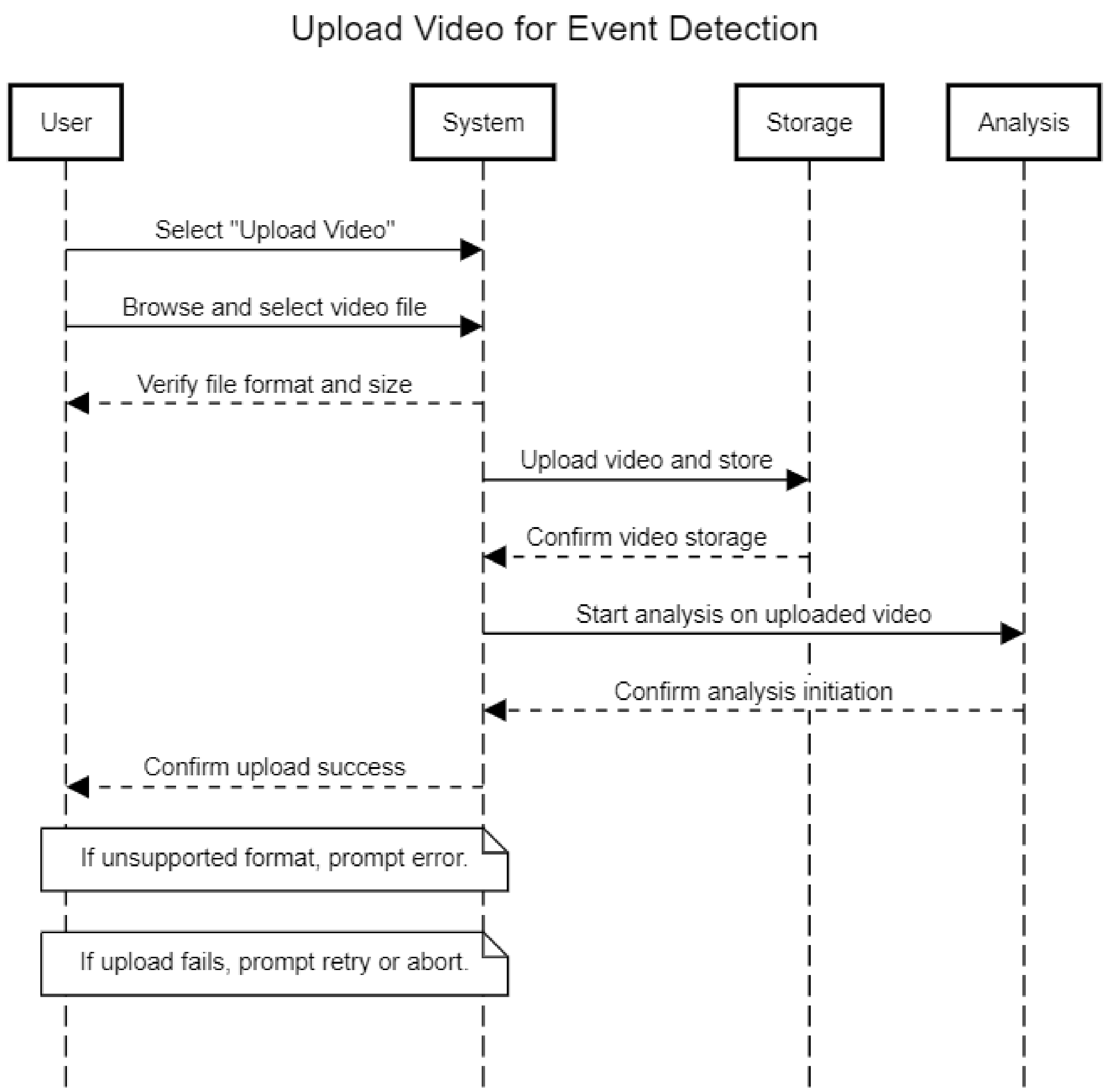

Figure 3.7: System Sequence Diagram for Upload Video for Event Detection

3.2.3.2 Generate Summarized Video

The SSD for the "Generate Summarized Video" use case depicts the steps involved in generating a video summary after processing. The system receives a request to generate a summary based on detected events and responds with the compiled video.

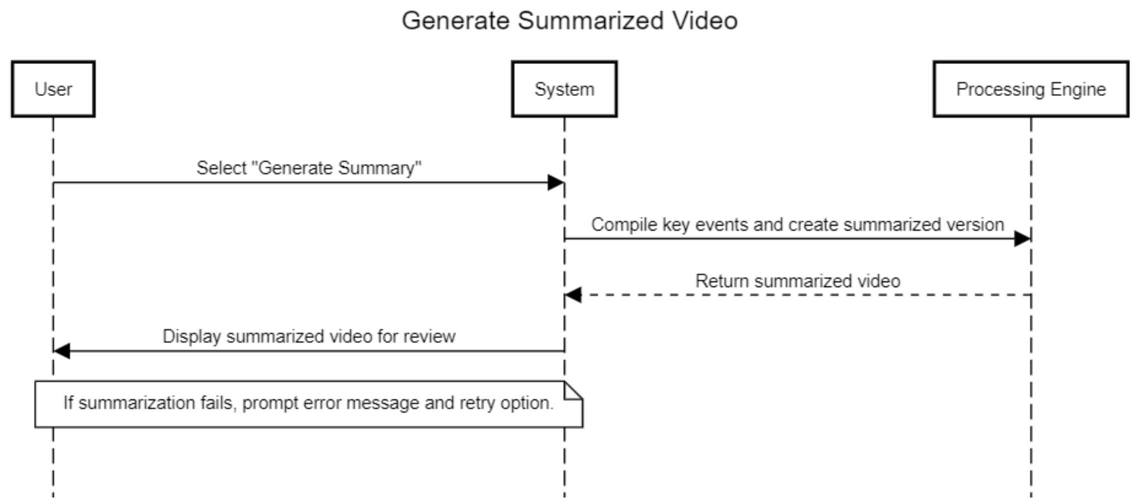

Figure 3.8: System Sequence Diagram for Generate Summarized Video

3.2.4 State Transition Diagram

The State Transition Diagram shows the various states the system undergoes, especially during event handling and backend processing. This diagram represents how the system transitions between states like “Idle,” “Processing,” and “Summarizing” based on the events detected in the video.

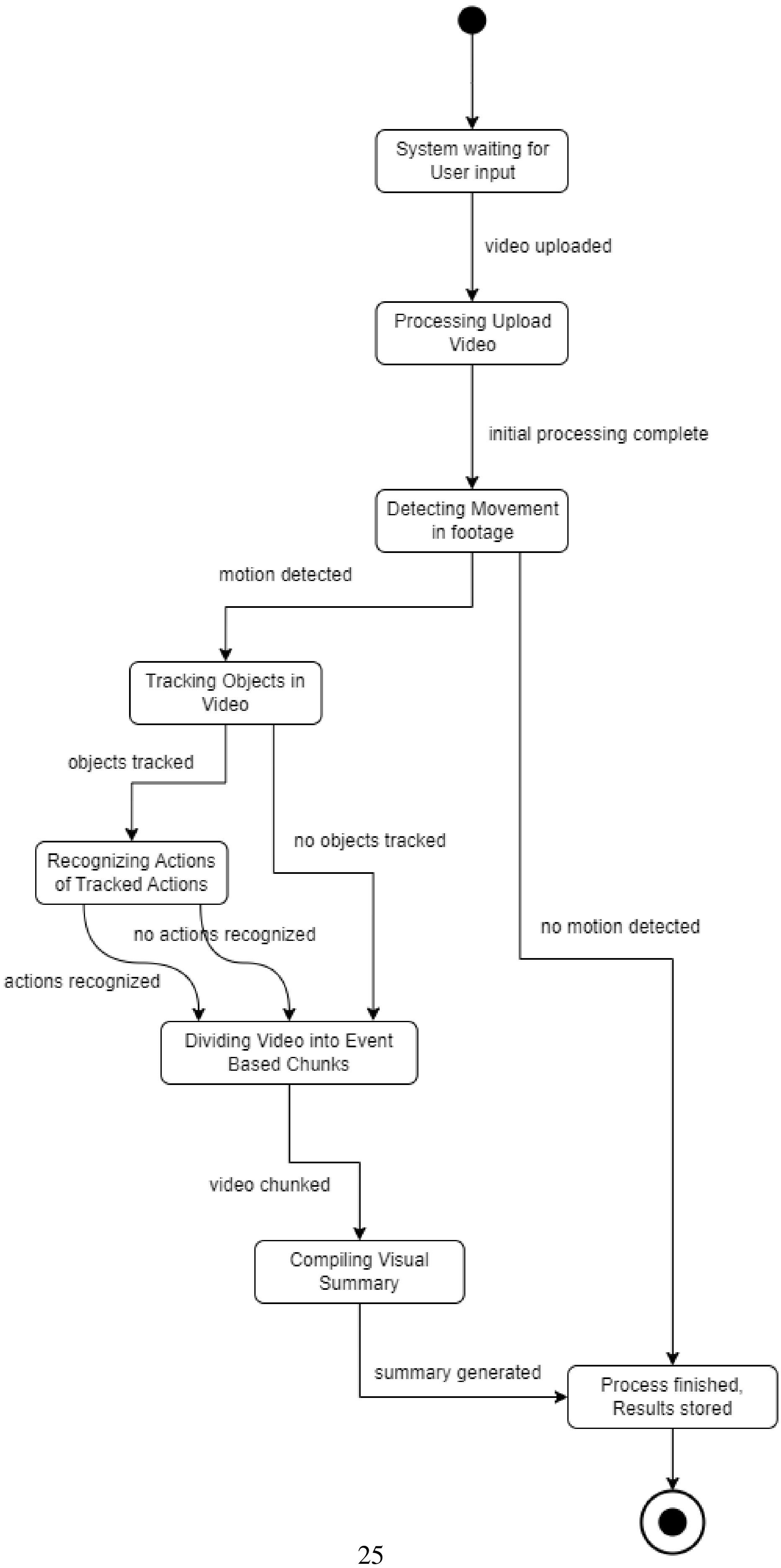

Figure 3.9: State Transition Diagram: Video Processing and Event Handling

3.3 Data Design

This section explains how the information domain of the system is transformed into data structures and how the major data or system entities are stored, processed, and organized.

3.3.1 Data Structures Overview

The data in the system is organized around key entities such as **User**, **Video**, **Event**, **Object**, **Action**, and **Summary**. These entities will be stored, processed, and managed efficiently in various data structures to support system functionality.

3.3.1.1 1. User Data

- **Attributes**: UserID, Username, Email, Password
- **Storage**: This data is stored in a relational database, with authentication details secured using encryption (e.g., hashed passwords).
- **Processing**: Data is used for authentication and authorization when users upload videos or access video summaries.

3.3.1.2 Video Data

- **Attributes**: VideoID, Filename, Duration, Resolution, UploadTime, UserID (Foreign Key)
- **Storage**: Videos are stored in cloud storage or local servers, while metadata is stored in a relational database. A file storage system (e.g., AWS S3, Google Cloud Storage) is used for video files.
- **Processing**: The video processing module fetches and processes videos for event detection and object tracking.

3.3.1.3 Event Data

- **Attributes**: EventID, EventType, Timestamp, VideoID (Foreign Key)
- **Storage**: Detected events are stored with timestamps and linked to the respective video in the relational database.
- **Processing**: Events are generated during video processing, triggering further actions such as action recognition.

3.3.1.4 Object Data

- **Attributes**: ObjectID, ObjectType, VideoID (Foreign Key)
- **Storage**: Detected objects (e.g., people, animals) are stored in a database and linked to the respective video.
- **Processing**: The system tracks the movements of these objects across the video footage.

3.3.1.5 Action Data

- **Attributes**: ActionID, ActionType, Timestamp, ObjectID (Foreign Key)
- **Storage**: Actions performed by objects (e.g., walking, running) are stored with timestamps and linked to objects.
- **Processing**: These actions are analyzed and included in event-based reports.

3.3.1.6 Timestamp Data

- **Attributes**: Timestamp, EventID (Foreign Key), ActionID (Foreign Key)
- **Storage**: Timestamps are used to mark key moments in the video and are stored in the database.
- **Processing**: Time-based navigation is enabled for the review of key events and actions.

3.3.1.7 Summary Data

- **Attributes**: SummaryID, VideoID (Foreign Key), SummaryText
- **Storage**: Summaries are stored as text or JSON in the database, associated with the analyzed video.
- **Processing**: Summaries provide a concise report of detected events, objects, and actions.

3.3.2 Databases and Storage Solutions

- **MongoDB Database**:

  - MongoDB is used to store unstructured and semi-structured data like **User**, **Video**, **Event**, **Object**, **Action**, **Timestamp**, and **Summary** as documents in collections.

  - Provides flexibility in handling different types of video analysis data without rigid schema constraints, making it ideal for dynamic and evolving datasets.

  - Enables efficient querying through its document-based model, supporting complex queries for event detection and reporting.

3.3.3 Data Flow and Processing

- When a user uploads a video, the metadata is stored in a relational database, and the video is stored in the file storage system.

- The **Video Processing Module** extracts key frames and processes the video to detect **Events** and **Objects**. These detections are stored in the database.

- The **Summarization Module** compiles the data into a report that users can access for further review.

3.3.4 Entity-Relationship Diagram (ERD)

The Entity-Relationship Diagram (ERD) visually represents the key entities and their relationships in the system.

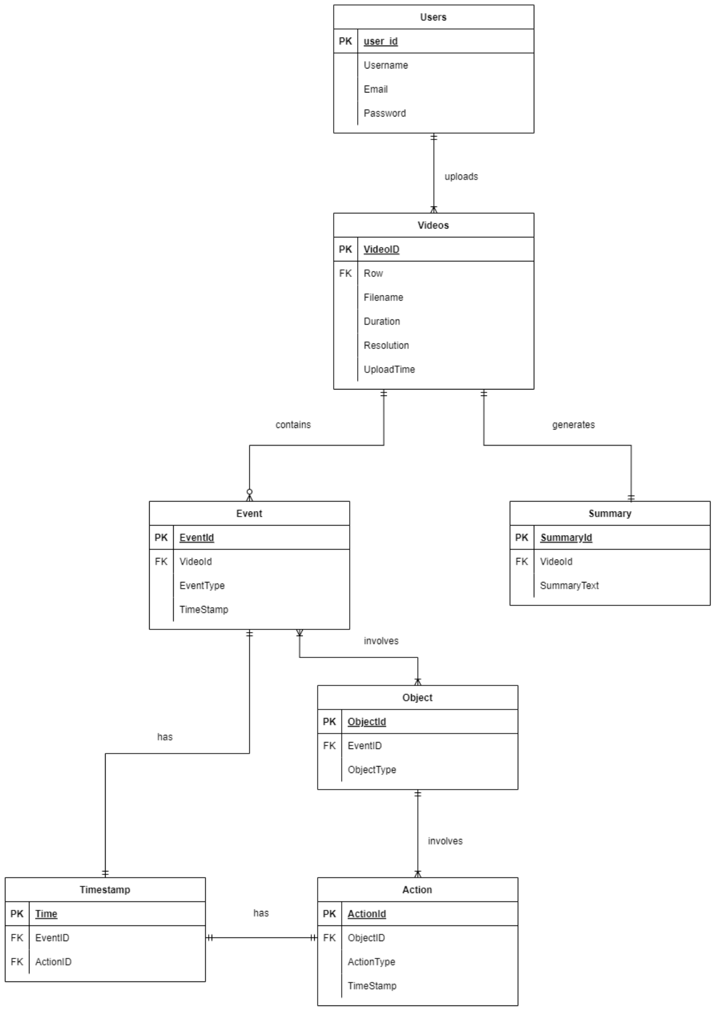

Figure 3.10: Entity-Relationship Diagram (ERD) of the System

In summary, the system uses a combination of structured and unstructured data storage to handle large volumes of video data efficiently, while the relational database ensures that key metadata and events are easily accessible and can be processed for analysis and reporting.

Bibliography

[1] Abdolamir Karbalaie, Farhad Abtahi, and Märten Sjöström. Event detection in surveillance videos: a review. *Multimedia Tools and Applications*, 81:35463–35501, 2022. Open access.